

## RESEARCH ARTICLE

# Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Email: corresponding@institution.edu**Abstract**

The ICH E17 guideline recommends regional pooling in multi-regional clinical trials (MRCTs) based on similarity of effect modifier (EM) distributions, but provides no specific methodology for quantifying such similarity. Existing approaches focus on location differences (standardized mean difference) or lack interpretable scales (Kolmogorov–Smirnov statistic). We propose the normalized Area Between Cumulative Distributions (nABCD), defined as the Wasserstein-1 distance between two distributions divided by twice the pooled interquartile range. Through its connection to the treatment effect heterogeneity bound, nABCD enables clinical calibration: the maximum potential regional treatment effect difference ( $\Delta_{\max}$ ) can be expressed as a function of nABCD, the pooled IQR, and the CATE sensitivity of the EM, with bootstrap confidence intervals quantifying estimation uncertainty. Simulation studies across eight scenarios—including location shifts, scale differences, skewed distributions, and combined patterns—at sample sizes typical in MRCT regional subgroups demonstrated satisfactory bias and coverage of the percentile bootstrap estimator. Unlike the standardized mean difference, nABCD captured variance, shape, and skewness differences that are invisible to location-only metrics. Application to a diabetes MRCT illustrated that clinical calibration provides context-dependent interpretation: a large distributional difference in a weak effect modifier (BMI: nABCD = 0.51,  $\Delta_{\max}$  = 0.16%) carried less clinical weight than a moderate difference in a strong effect modifier (baseline HbA1c: nABCD = 0.27,  $\Delta_{\max}$  = 0.24%), demonstrating that distributional magnitude and clinical impact are fundamentally different dimensions. The nABCD framework fills a methodological gap in ICH E17 implementation by translating distributional differences into potential treatment effect heterogeneity on the clinical scale, supporting evidence-based pooling decisions.

**KEYWORDS**

multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

## 1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.<sup>1,2</sup> This approach offers substantial benefits: accelerated timelines, broader generalizability, and earlier access to new therapies for patients worldwide. The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs, with a central assumption that treatment effects are generalizable across the target population.<sup>3</sup>

A key strategy for enhancing the ability to assess regional consistency in treatment outcomes is the regional pooling approach, wherein regions with similar patient characteristics are grouped for analysis.<sup>3</sup> The ICH E17 guideline explicitly recommends that pooling decisions be based on the similarity of effect modifier (EM) distributions:

**Abbreviations:** ABCD, area between cumulative distributions; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; CKD, chronic kidney disease; EM, effect modifier; EU, European Union; ICH, International Council for Harmonisation; IQR, interquartile range; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; nABCD, normalized area between cumulative distributions; NMPA, National Medical Products Administration; SMD, standardized mean difference; US, United States.

“Regions may be pooled for randomisation and/or analysis if subjects are thought to be *similar enough* with respect to intrinsic and/or extrinsic factors relevant to the disease and/or drug under study.” (ICH E17, Section 2.2.5)

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. For example, if younger patients respond better to treatment than older patients, age is an effect modifier. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why EM distributional similarity is critical to the validity of regional pooling.

Despite the regulatory importance of EM distributional similarity, current practice lacks a standardized quantitative methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are “similar enough.” Recent regulatory guidance has highlighted this gap. Song et al., writing from the China NMPA perspective on ICH E17 implementation, note the challenge of operationalizing pooling criteria without quantitative tools.<sup>4</sup> Long et al. further discuss basic considerations for consistency evaluation under ICH E17.<sup>5</sup>

Current approaches to assessing distributional similarity have significant limitations (Table 1). The standardized mean difference, while widely used for baseline covariate comparisons,<sup>6</sup> fundamentally cannot detect differences in variance or distributional shape—precisely the types of differences that may drive treatment effect heterogeneity through effect modification.

**TABLE 1** Limitations of current approaches to distributional similarity assessment.

| Method                             | Limitation                                      |
|------------------------------------|-------------------------------------------------|
| Visual inspection                  | Subjective, not reproducible                    |
| Standardized mean difference (SMD) | Captures only location, ignores scale and shape |
| Kolmogorov–Smirnov statistic       | No interpretable scale for decision-making      |

This paper addresses the methodological gap by proposing the *normalized Area Between Cumulative Distributions* (nABCD), a metric for comparing EM distributions across regions, together with a clinical calibration framework that translates distributional differences into potential treatment effect heterogeneity on the outcome scale. Our specific research question is:

How can we estimate distributional similarity between regions in a scale-free manner, and translate that estimate into clinically interpretable information about potential treatment effect heterogeneity?

The emphasis is on *estimation and clinical interpretation*, not hypothesis testing. We seek to provide regulatory scientists with quantitative tools that inform deliberation, not with binary accept/reject rules. This design philosophy reflects the ICH E17 principle that pooling decisions require contextual judgment rather than mechanical application of thresholds.

The nABCD metric measures the total area between two cumulative distribution functions, normalized by the pooled interquartile range to achieve scale-free interpretation. The framework offers four contributions:

1. *Full distributional comparison*: The Wasserstein-1 distance captures differences in location, scale, and shape simultaneously.<sup>7</sup>
2. *Scale-free estimation*: Normalization by IQR yields a unitless measure, enabling comparisons across EMs measured on different scales. Bootstrap confidence intervals quantify estimation uncertainty.
3. *Clinical calibration*: Through the heterogeneity bound (Proposition 2), nABCD estimates can be translated into potential treatment effect differences ( $\Delta_{\max}$ ) specific to each EM’s clinical context, expressed on the primary outcome scale.<sup>8</sup>
4. *Sensitivity analysis*: Because the calibration depends on the CATE sensitivity parameter  $L$ , which may be uncertain, the framework naturally accommodates sensitivity analysis over  $L$ , providing a richer evidence base for decision-making than a single binary test.

We focus on continuous effect modifiers, for which the first Wasserstein distance provides a natural and theoretically grounded measure of distributional dissimilarity. Categorical or mixed-type effect modifiers (e.g., genotype, race, prior treatment history) require alternative distance measures and are beyond the scope of this work.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework. Section 3 describes a comprehensive simulation study. Section 4 illustrates application to an MRCT dataset. Section 5 discusses implications, limitations, and future directions.

## 2 | METHODS

### 2.1 | The nABCD Metric

An effect modifier (EM) is a baseline patient characteristic for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted  $\tau(x) = E[Y(1) - Y(0) | X = x]$ , where  $X$  is the effect modifier. When this function is non-constant, the average treatment effect observed in region  $r$  depends on the distribution  $F_r$  of the EM in that region:

$$\bar{\tau}_r = \int \tau(x) dF_r(x). \quad (1)$$

To formalize the relationship between distributional differences and treatment effect heterogeneity, let the CATE function be bounded with Lipschitz constant  $L$ . The difference in regional average treatment effects can be bounded by:

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2), \quad (2)$$

where  $W_1(F_1, F_2)$  denotes the Wasserstein-1 distance between EM distributions in regions 1 and 2.

The Wasserstein-1 distance (also known as the Earth Mover's Distance) between two cumulative distribution functions  $F$  and  $G$  is defined as:<sup>9</sup>

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx. \quad (3)$$

Geometrically, this equals the total area between the two CDFs (Figure 1). Unlike the standardized mean difference, the Wasserstein distance responds to changes in variance, skewness, and other distributional features.<sup>7</sup> We use  $W_1$  rather than higher-order Wasserstein distances because the Lipschitz bound in equation (2) requires the  $W_1$  distance specifically. By the Kantorovich–Rubinstein duality,  $W_1(F_1, F_2) = \sup_{\|f\|_{\text{Lip}} \leq 1} |\int f dF_1 - \int f dF_2|$ , which directly connects Lipschitz-continuous functions (including the CATE) to distributional distance.<sup>9</sup> The  $W_2$  distance, while admitting closed-form expressions for Gaussian families, does not possess this dual characterization via Lipschitz functions and therefore cannot provide the heterogeneity bound that is central to our clinical calibration framework.

The *normalized Area Between Cumulative Distributions* (nABCD) is defined as the Wasserstein-1 distance normalized by twice the pooled interquartile range:

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot \text{IQR}_{\text{pooled}}}, \quad (4)$$

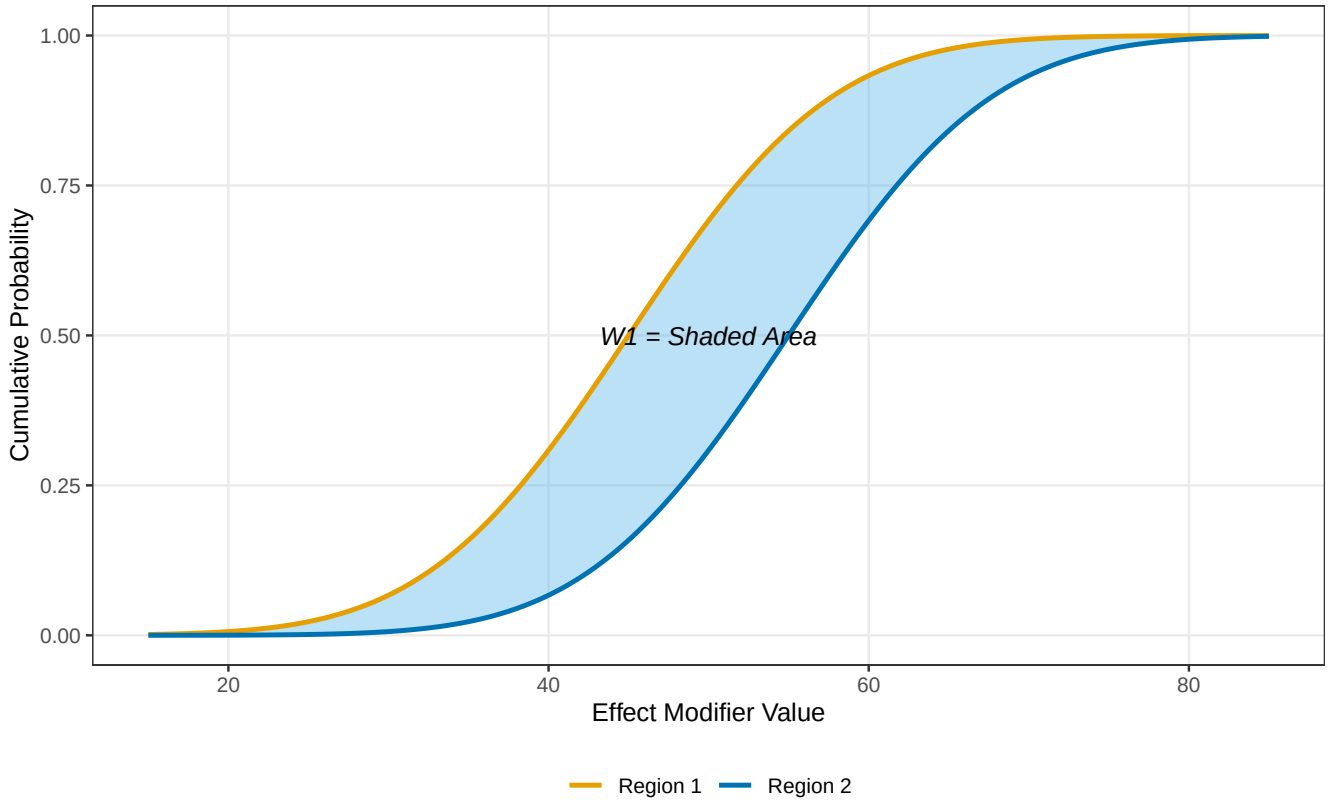
where the pooled IQR is computed from the combined sample. The IQR-based normalization enables scale-free interpretation, is resistant to outliers, and expresses distributional differences in units of spread. We chose the IQR over alternative scale estimators such as the standard deviation or the  $Q_n$  statistic of Rousseeuw and Croux.<sup>10</sup> The IQR is directly interpretable as the central 50% spread of the EM distribution, a quantity familiar to clinical researchers who routinely report medians and IQRs for non-normally distributed variables. Although  $Q_n$  offers a higher breakdown point (50% vs. 25%) and greater Gaussian efficiency, the high breakdown point is unnecessary here: EM distributions are population-level quantities, not contaminated measurement data, so the robustness advantage of  $Q_n$  does not materialize in this setting.

**Proposition 1.** (Non-negativity). *For distributions with finite first moment, provided  $\text{IQR}_{\text{pooled}} > 0$  (i.e., the pooled distribution is non-degenerate),  $\text{nABCD} \geq 0$ , with equality if and only if  $F_1 = F_2$ .*

**Proposition 2.** (Connection to heterogeneity). *If the CATE function has Lipschitz constant  $L$ , then*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2). \quad (5)$$

*Proof.* Substituting (4) into (2) yields  $|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2) = L \cdot 2 \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2) = 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2)$ .  $\square$



**FIGURE 1** nABCD as the area between cumulative distribution functions. The shaded region represents the Wasserstein-1 distance  $W_1(F_1, F_2)$ , which equals the total area between the two CDFs. nABCD normalizes this area by twice the pooled IQR to achieve scale-free interpretation.

## 2.2 | Estimation

Given samples  $\{X_{1,i}\}_{i=1}^{n_1}$  from region 1 and  $\{X_{2,j}\}_{j=1}^{n_2}$  from region 2, nABCD is estimated using empirical distribution functions:

$$\widehat{\text{nABCD}} = \frac{\sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)})}{2 \cdot \widehat{\text{IQR}}_{\text{pooled}}}, \quad (6)$$

where  $x_{(1)} < \dots < x_{(n_1+n_2)}$  are the combined order statistics.

*Computational complexity:*  $O((n_1 + n_2) \log(n_1 + n_2))$ , dominated by sorting.

We employ the nonparametric percentile bootstrap for inference<sup>11,12</sup> with  $B = 2,000$  replicates. In one dimension,  $W_1$  equals the  $L_1$  distance between CDFs, for which del Barrio et al.<sup>11</sup> established  $\sqrt{n}$ -convergence to a functional of the Brownian bridge; the two-sample extension follows by standard arguments with rate  $\sqrt{n_1 n_2 / (n_1 + n_2)}$ . When  $F_1 \neq F_2$ , the Hadamard derivative of the  $L_1$  functional is linear—the sign of  $F_1(x) - F_2(x)$  is constant almost everywhere—so the ordinary nonparametric bootstrap is consistent.<sup>12</sup> Near the boundary  $F_1 = F_2$ , the derivative becomes non-linear (a supremum over a convex set), and the naive bootstrap may exhibit modest undercoverage, consistent with the finite-sample behaviour observed in our null-adjacent scenarios (Section 3).

## 2.3 | Interpretation and Clinical Calibration

A central feature of nABCD is its connection to treatment effect heterogeneity through Proposition 2. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed thresholds alone.

### 2.3.1 | Clinical Calibration via the Heterogeneity Bound

From equation (5), the maximum potential difference in regional average treatment effects attributable to EM distributional differences is:

$$\Delta_{\max} = 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}(F_1, F_2), \quad (7)$$

where  $L$  is the Lipschitz constant of the CATE function  $\tau(x)$  with respect to the EM, reflecting the clinical sensitivity of treatment effect to that EM. This use of  $L$  as a sensitivity parameter follows the framework proposed by Armstrong and Kolesár,<sup>13</sup> who recommended reporting confidence intervals for a range of plausible Lipschitz constants. For a given EM,  $L$  can be estimated from prior knowledge, pilot data, or published subgroup analyses.

The clinical calibration procedure is as follows:

1. Identify candidate EMs and compute nABCD with bootstrap CIs for each EM across region pairs.
2. For each EM, estimate or bound  $L$  from prior knowledge (e.g., subgroup analyses showing treatment effect varies by  $\Delta\tau$  per unit change in the EM, giving  $L \approx \Delta\tau/\Delta x$ ).
3. Compute  $\Delta_{\max}$  using equation (7). This represents the worst-case treatment effect difference attributable to the distributional difference.
4. Derive the bootstrap CI for  $\Delta_{\max}$  from the nABCD confidence interval:  $[\Delta_{\max,L}, \Delta_{\max,U}] = 2L \cdot \text{IQR}_{\text{pooled}} \cdot [\text{nABCD}_L, \text{nABCD}_U]$ . This expresses inferential uncertainty directly on the clinical scale.
5. Report  $\Delta_{\max}$  and its CI alongside the clinical context—for example, the overall treatment effect size, the non-inferiority margin, or the minimal clinically important difference. This information enables trial designers and regulatory scientists to exercise clinical judgment about pooling, considering the totality of evidence rather than a binary accept/reject decision.

This estimation-centered approach deliberately avoids forcing pooling decisions into a binary hypothesis testing framework. The rationale is threefold. First, the ICH E17 guideline describes “similar enough” as inherently context-dependent, and reducing this judgment to a rejection threshold risks oversimplification. Second, the uncertainty in  $L$  means that  $\Delta_{\max}$  itself is subject to sensitivity analysis (Section 4); a binary test cannot naturally accommodate this layered uncertainty. Third, consistent with recent calls to move beyond dichotomous statistical decisions,<sup>14</sup> presenting  $\Delta_{\max}$  with its CI provides richer information for regulatory deliberation than a p-value or a reject/fail-to-reject outcome.

When regulatory agencies require a formal decision rule, the estimation framework can be adapted: declare pooling acceptable if the upper bound of the 95% CI for  $\Delta_{\max}$  falls below a pre-specified clinical margin  $\Delta_{\text{clin}}$ . However, we recommend this as a supplementary rather than primary use of the nABCD framework.

### 2.3.2 | Reference Benchmarks

When prior knowledge of  $L$  is unavailable, the benchmarks in Table 2 provide a convenience reference for initial assessment. These labels describe *distributional magnitude* only and should not be conflated with clinical significance: as Section 4 demonstrates, a “large” nABCD for a weak effect modifier may carry less clinical consequence than a “moderate” nABCD for a strong effect modifier, because CATE sensitivity  $L$  determines how distributional distance translates into treatment effect heterogeneity.

**TABLE 2** Reference benchmarks for nABCD values when CATE sensitivity  $L$  is unknown.

| nABCD Range | Interpretation        | Suggested Action                  |
|-------------|-----------------------|-----------------------------------|
| < 0.05      | Negligible difference | Pooling broadly supportable       |
| 0.05–0.15   | Small difference      | Pooling generally acceptable      |
| 0.15–0.30   | Moderate difference   | Proceed with clinical calibration |
| > 0.30      | Large difference      | Clinical calibration essential    |

These benchmarks serve as initial reference points only; clinical interpretation should always proceed through the calibration framework of Section 2.3, as the same nABCD value may have different clinical implications depending on the Lipschitz constant  $L$ . The same nABCD value may be negligible for one EM and consequential for another depending on CATE sensitivity. Actual pooling decisions should use clinical calibration (equation 7) whenever possible.

### 3 | SIMULATION STUDY

We conducted simulation studies to evaluate the estimation properties of the nABCD estimator: bias, variability, and coverage probability of bootstrap confidence intervals. These properties are essential for the clinical calibration framework, as reliable estimation of nABCD directly determines the quality of  $\Delta_{\max}$  estimates. We assessed performance across a range of scenarios relevant to MRCT applications.

#### 3.1 | Simulation Design

##### 3.1.1 | Scenarios

We designed eight systematic scenarios for methodological validation, examining controlled distributional differences across location, scale, shape, and combinations thereof. Table 3 summarizes these scenarios with their true nABCD values.

**TABLE 3** Simulation scenarios with clinical motivations.

| ID | Description          | Distribution 1 | Distribution 2             | True nABCD |
|----|----------------------|----------------|----------------------------|------------|
| S1 | Null (identical)     | $N(50, 10^2)$  | $N(50, 10^2)$              | 0.000      |
| S2 | Location $0.2\sigma$ | $N(50, 10^2)$  | $N(52, 10^2)$              | 0.074      |
| S3 | Location $0.5\sigma$ | $N(50, 10^2)$  | $N(55, 10^2)$              | 0.186      |
| S4 | Location $1.0\sigma$ | $N(50, 10^2)$  | $N(60, 10^2)$              | 0.372      |
| S5 | Scale $1.5\times$    | $N(50, 10^2)$  | $N(50, 15^2)$              | 0.148      |
| S6 | Shape (Gamma)        | $N(50, 10^2)$  | $\text{Gamma}(25, 0.5)$    | 0.067      |
| S7 | Skew (Log-normal)    | $N(50, 10^2)$  | $\text{LogN}(\mu, \sigma)$ | 0.302      |
| S8 | Location + Scale     | $N(50, 10^2)$  | $N(55, 15^2)$              | 0.175      |

The Gamma distribution in S6 has shape parameter 25 and rate 0.5, yielding mean 50 and standard deviation 10. The Log-normal in S7 is parameterized to match mean 50 with CV  $\approx 53\%$ , typical of clinical biomarkers such as ALT. S8 combines location and scale differences, representing the most realistic MRCT pattern.

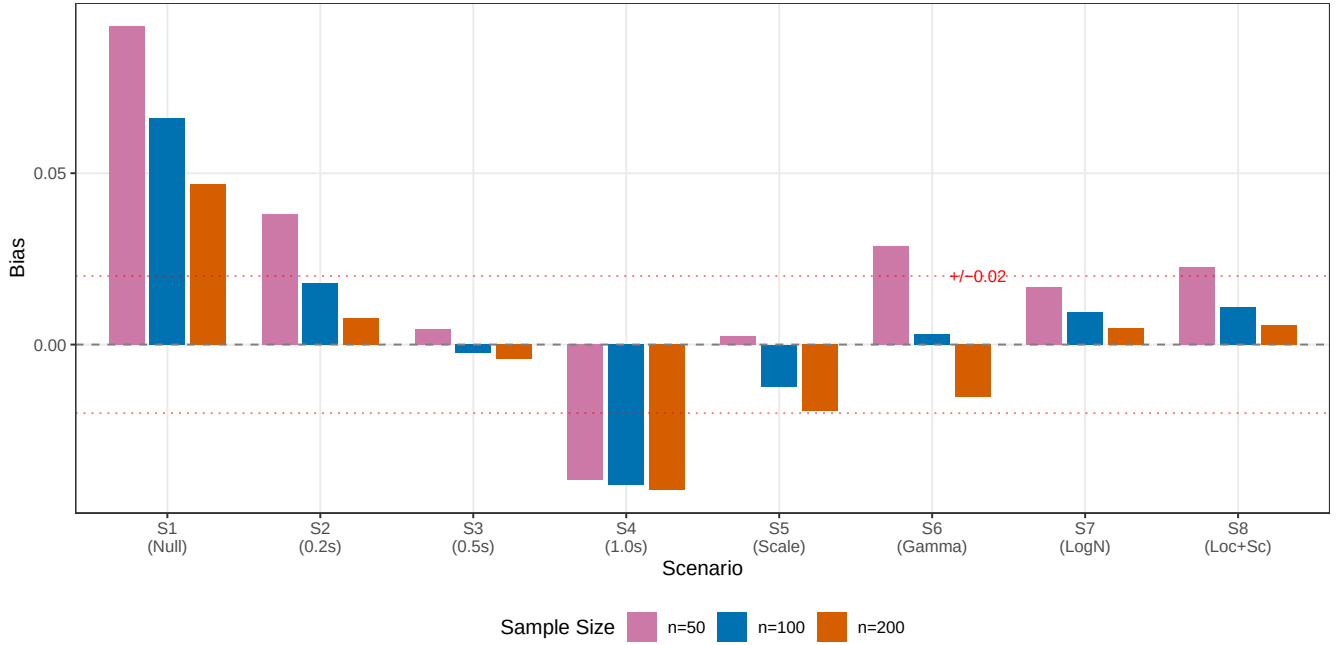
##### 3.1.2 | Simulation Parameters

For each scenario, we generated samples of size  $n = 50, 100$ , and  $200$  per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using  $B = 2,000$  resamples. All simulations were conducted in R version 4.3.3.

##### 3.1.3 | Evaluation Metrics

Consistent with the estimation-centered framework, we evaluated:

1. *Bias*:  $\text{Mean}(\widehat{\text{nABCD}}) - \text{nABCD}$
2. *Root mean squared error (RMSE)*:  $\sqrt{\text{Mean}[(\widehat{\text{nABCD}} - \text{nABCD})^2]}$
3. *Coverage probability*: Proportion of 95% bootstrap CIs containing the true value
4. *CI width*: Mean width of bootstrap CIs, reflecting estimation precision



**FIGURE 2** Bias of nABCD estimator by scenario and sample size. Horizontal dashed lines indicate  $\pm 0.02$  bias threshold. For most scenarios, bias is less than 0.02 at  $n \geq 100$ , with S4 being the exception due to boundary effects.

## 3.2 | Results

### 3.2.1 | Point Estimation and Coverage

Table 4 presents the bias of the nABCD estimator across scenarios and sample sizes. The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 0.093 at  $n = 50$  to 0.047 at  $n = 200$ . This positive bias is attributable to the non-negative nature of the Wasserstein distance: even when true nABCD equals zero, sampling variability produces positive estimates.

**TABLE 4** Bias of nABCD estimator by scenario and sample size.

| Scenario           | True nABCD | $n = 50$ | $n = 100$ | $n = 200$ |
|--------------------|------------|----------|-----------|-----------|
| S1 (Null)          | 0.000      | 0.093    | 0.066     | 0.047     |
| S2 (0.2 $\sigma$ ) | 0.074      | 0.038    | 0.018     | 0.007     |
| S3 (0.5 $\sigma$ ) | 0.186      | 0.004    | -0.002    | -0.005    |
| S4 (1.0 $\sigma$ ) | 0.372      | -0.039   | -0.041    | -0.042    |
| S5 (Scale)         | 0.148      | 0.002    | -0.012    | -0.019    |
| S6 (Shape)         | 0.067      | 0.028    | 0.003     | -0.015    |
| S7 (Skew)          | 0.302      | 0.019    | 0.009     | 0.005     |
| S8 (Loc+Scale)     | 0.175      | 0.024    | 0.011     | 0.006     |

Bias is defined as  $\text{Mean}(\widehat{\text{nABCD}}) - \text{nABCD}$ . Results based on 10,000 replications with  $B = 2,000$  bootstrap resamples.

For most non-null scenarios, bias was less than 0.02 in absolute value at  $n \geq 100$ , indicating satisfactory point estimation performance at practical sample sizes. For S4 (1.0 $\sigma$  location shift), negative bias of approximately  $-0.04$  persisted across all sample sizes, reflecting the bounded nature of nABCD near its theoretical upper range (Figure 2). The new scenarios S7 (log-normal) and S8 (location + scale) showed small positive bias that diminished with increasing sample size, following the expected pattern for mid-range true nABCD values.

Table 5 presents coverage probabilities of the 95% bootstrap confidence intervals. For  $n \geq 100$ , coverage was within 0.87–0.98 for most scenarios, with S2 showing moderate undercoverage at  $n = 100$  (0.896). S4 showed progressive undercoverage with increasing  $n$  (0.874 at  $n = 100$ , 0.740 at  $n = 200$ ), attributable to the persistent negative bias under large distributional differences. At  $n = 50$ , undercoverage was more pronounced, particularly for S6 (0.576) and S2 (0.669), indicating that this sample size is insufficient for reliable inference.

Coverage exhibited a non-monotonic pattern across sample sizes for the shape scenario (S6): 0.576 at  $n = 50$ , 0.949 at  $n = 100$ , and 0.997 at  $n = 200$ . The low coverage at  $n = 50$  reflects large positive bias relative to CI width, while the overcoverage at  $n = 200$  reflects the combination of shrinking bias ( $-0.015$ ) with CIs that remain relatively wide for this small true value. The new scenarios S7 (skew) and S8 (location + scale) showed excellent coverage: S7 achieved near-nominal coverage across all sample sizes (0.951–0.954), while S8 showed slight undercoverage that improved with sample size (0.916 at  $n = 50$  to 0.939 at  $n = 200$ ).

We also evaluated BCa (bias-corrected and accelerated) confidence intervals. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, with the largest discrepancy in scale and shape scenarios (e.g., S5 at  $n = 100$ : percentile 0.979 vs. BCa 0.841; S6 at  $n = 200$ : percentile 0.997 vs. BCa 0.493). The BCa correction overcorrects for nABCD because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

**TABLE 5** Coverage probability of 95% percentile bootstrap confidence intervals.

| Scenario           | $n = 50$ | $n = 100$ | $n = 200$ |
|--------------------|----------|-----------|-----------|
| S2 (0.2 $\sigma$ ) | 0.669    | 0.896     | 0.943     |
| S3 (0.5 $\sigma$ ) | 0.950    | 0.949     | 0.951     |
| S4 (1.0 $\sigma$ ) | 0.928    | 0.874     | 0.740     |
| S5 (Scale)         | 0.962    | 0.979     | 0.933     |
| S6 (Shape)         | 0.576    | 0.949     | 0.997     |
| S7 (Skew)          | 0.953    | 0.954     | 0.951     |
| S8 (Loc+Scale)     | 0.916    | 0.932     | 0.939     |

Coverage under S1 (Null) is not reported because the true value of 0 lies at the boundary of the parameter space.

For S4 (large effect), coverage decreased at larger sample sizes due to increased precision revealing the persistent negative bias. The modest coverage degradation at small nABCD values (near the null  $F_1 = F_2$ ) is consistent with known theoretical properties of bootstrap inference for Wasserstein distances: at  $F_1 = F_2$ , the Hadamard derivative of  $W_1$  becomes non-linear, which can affect bootstrap consistency.<sup>12</sup> This boundary behavior diminishes with increasing sample size and increasing distributional separation, as confirmed by the near-nominal coverage observed for S3–S8 at  $n \geq 100$ .

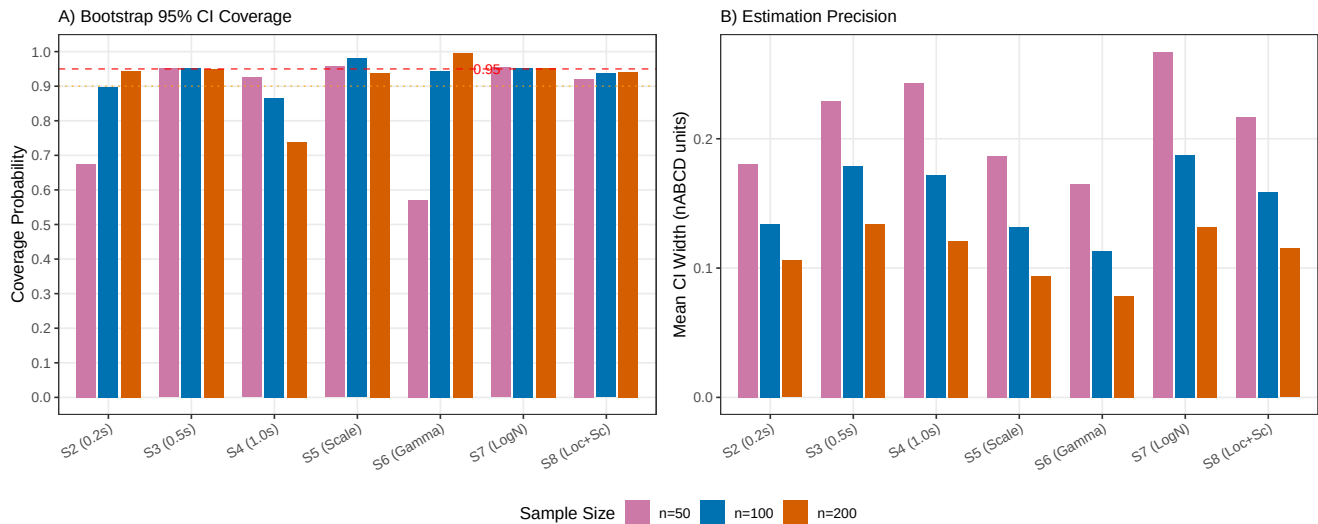
### 3.2.2 | Estimation Precision

Figure 3 summarizes the estimation quality across scenarios and sample sizes, presenting both coverage probabilities and mean CI widths.

Table 6 presents the RMSE and mean CI width across scenarios. RMSE decreased consistently with increasing  $n$ , reaching values below 0.06 for all scenarios at  $n = 200$ . Mean CI width narrowed proportionally, with CIs at  $n = 100$  typically spanning 0.11–0.19 in nABCD units. In the clinical calibration framework, this CI width propagates to  $\Delta_{\max}$ : for example, with  $L = 0.3$  and  $IQR = 1.5$ , a CI width of 0.13 in nABCD corresponds to a CI width of  $2 \times 0.3 \times 1.5 \times 0.13 = 0.12\%$  HbA1c—a level of precision sufficient for meaningful clinical calibration.

Under the null hypothesis (S1), the positive bias produces CIs that are shifted upward from zero. At  $n = 50$ , the mean estimate was 0.093 with mean CI width 0.16; at  $n = 200$ , the mean estimate decreased to 0.047 with mean CI width 0.08. This bias is relevant for clinical calibration: when the true distributional difference is zero, the estimator will suggest a small positive  $\Delta_{\max}$ . Practitioners should be aware of this conservative tendency, particularly at smaller sample sizes.





**FIGURE 3** Estimation quality of nABCD bootstrap inference. Panel (A) shows coverage probability of 95% bootstrap confidence intervals; dashed and dotted lines indicate the nominal 0.95 and minimum acceptable 0.90 levels. Panel (B) shows mean CI width in nABCD units. Coverage exceeds 0.90 for most scenarios at  $n \geq 100$ ; CI width narrows consistently with increasing  $n$ .

**TABLE 6** RMSE and mean CI width by scenario and sample size.

| Scenario           | RMSE     |           |           | Mean CI Width |           |           |
|--------------------|----------|-----------|-----------|---------------|-----------|-----------|
|                    | $n = 50$ | $n = 100$ | $n = 200$ | $n = 50$      | $n = 100$ | $n = 200$ |
| S1 (Null)          | 0.100    | 0.070     | 0.050     | 0.16          | 0.11      | 0.08      |
| S2 (0.2 $\sigma$ ) | 0.060    | 0.043     | 0.032     | 0.18          | 0.13      | 0.11      |
| S3 (0.5 $\sigma$ ) | 0.066    | 0.049     | 0.035     | 0.23          | 0.18      | 0.13      |
| S4 (1.0 $\sigma$ ) | 0.073    | 0.060     | 0.052     | 0.24          | 0.17      | 0.12      |
| S5 (Scale)         | 0.045    | 0.035     | 0.031     | 0.19          | 0.13      | 0.09      |
| S6 (Shape)         | 0.045    | 0.024     | 0.023     | 0.16          | 0.11      | 0.08      |
| S7 (Skew)          | 0.070    | 0.048     | 0.034     | 0.27          | 0.19      | 0.13      |
| S8 (Loc+Scale)     | 0.062    | 0.043     | 0.031     | 0.22          | 0.16      | 0.12      |

CI width is defined as mean of (upper – lower) across 10,000 replications.

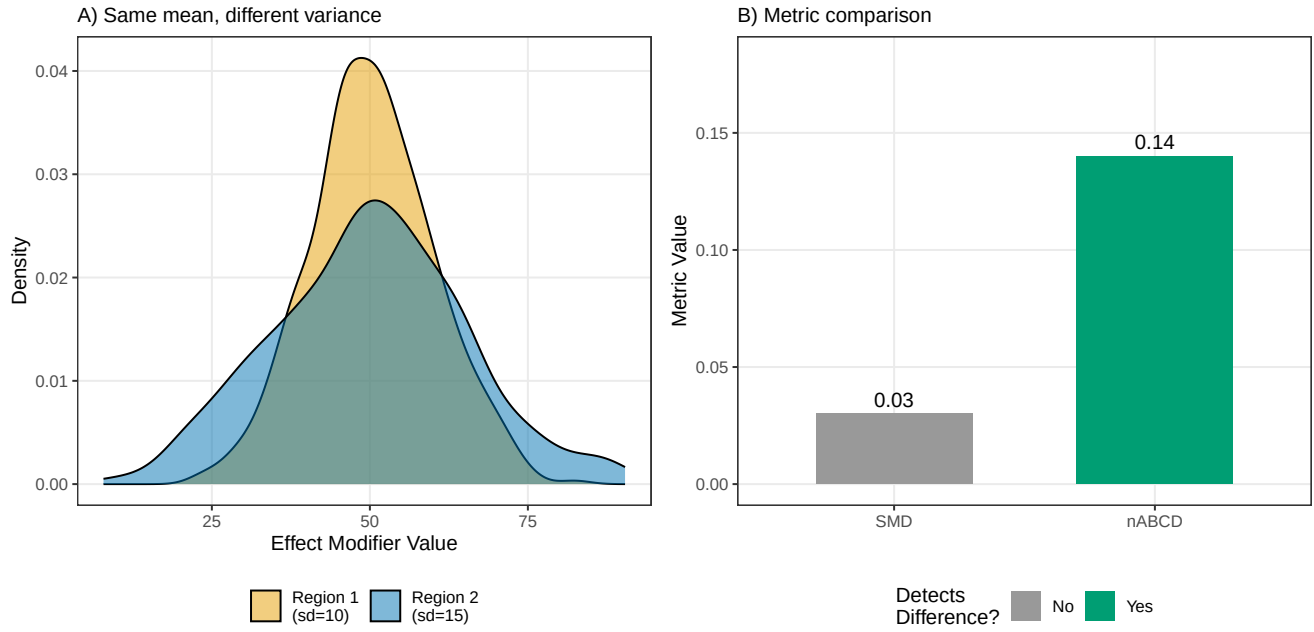
### 3.2.3 | Comparison with Standardized Mean Difference

Table 7 compares the sensitivity of nABCD and SMD to different types of distributional differences. While both metrics respond to location shifts, SMD is insensitive to differences in variance and shape—the very types of distributional differences that may drive treatment effect heterogeneity through non-linear CATE functions.

**TABLE 7** Sensitivity comparison: nABCD versus SMD at  $n = 100$ .

| Scenario        | nABCD (mean $\pm$ SD) | SMD (mean $\pm$ SD) | Implication        |
|-----------------|-----------------------|---------------------|--------------------|
| S3 (Location)   | 0.184 $\pm$ 0.049     | 0.50 $\pm$ 0.14     | Both detect        |
| S5 (Scale only) | 0.136 $\pm$ 0.033     | 0.00 $\pm$ 0.14     | Only nABCD detects |
| S6 (Shape only) | 0.070 $\pm$ 0.024     | 0.00 $\pm$ 0.14     | Only nABCD detects |
| S7 (Skew only)  | 0.311 $\pm$ 0.047     | 0.00 $\pm$ 0.14     | Only nABCD detects |

SMD depends only on the difference in means, making it insensitive to differences in variance, shape, and skewness. nABCD captures the full distributional difference. S7 is particularly informative: a large distributional difference (nABCD = 0.31) is entirely invisible to SMD.



**FIGURE 4** Comparison of nABCD and SMD across distributional difference types. nABCD responds to scale (S5) and shape (S6) differences where SMD remains near zero, demonstrating its advantage for full distributional comparison.

In summary, the simulation study across eight scenarios demonstrates that the nABCD estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. *Bias*: Less than 0.02 for most non-null scenarios at  $n \geq 100$ , with S4 showing persistent bias of  $-0.04$  due to boundary effects at large true values
2. *Coverage*: 0.87–0.98 at  $n \geq 100$  for most scenarios; new scenarios S7 (skew) and S8 (location + scale) achieved coverage between 0.93–0.95
3. *Precision*: RMSE below 0.06 and CI width below 0.19 at  $n \geq 100$
4. *Sensitivity*: Captures location, scale, and shape differences where SMD is limited to location only

Scenario S3 ( $0.5\sigma$  location shift, true nABCD = 0.186) exemplifies the operating characteristics at clinically relevant effect sizes: bias was negligible ( $-0.002$  at  $n = 100$ ), coverage was nominal (0.949), and CI width (0.18) translated to a  $\Delta_{\max}$  CI width of  $2 \times 0.3 \times 1.5 \times 0.18 = 0.16\%$  HbA1c when calibrated with  $L = 0.3$  and  $IQR = 1.5$ —a level of precision sufficient for informed regulatory deliberation.

These estimation properties ensure that nABCD, when combined with the clinical calibration framework, provides  $\Delta_{\max}$  estimates of sufficient quality for regulatory deliberation. We recommend  $n \geq 100$  per region for reliable estimation and inference.

## 4 | APPLICATION

### 4.1 | Example: Type 2 Diabetes MRCT

We illustrate the nABCD clinical calibration framework using a hypothetical MRCT in type 2 diabetes with three regions: Japan ( $n = 150$ ), United States ( $n = 200$ ), and European Union ( $n = 180$ ). The primary endpoint was change in HbA1c (%) at 24 weeks, with an overall treatment effect of  $-0.8\%$  and a pre-specified non-inferiority margin of  $\Delta_{\text{clin}} = 0.4\%$  HbA1c. This example directly addresses the ICH E17 guideline's emphasis on understanding regional differences in intrinsic and extrinsic ethnic

factors:<sup>3</sup> nABCD quantifies the distributional similarity of such factors across regions, providing the quantitative foundation that ICH E17 calls for but does not prescribe. Table 8 presents baseline characteristics by region.

**TABLE 8** Baseline characteristics by region in the hypothetical diabetes MRCT.

| Characteristic   | Japan ( $n = 150$ ) | US ( $n = 200$ ) | EU ( $n = 180$ ) |
|------------------|---------------------|------------------|------------------|
| Age, mean (SD)   | 62.3 (10.2)         | 58.7 (11.5)      | 60.1 (10.8)      |
| BMI, mean (SD)   | 24.8 (3.2)          | 32.1 (5.8)       | 29.4 (4.9)       |
| HbA1c, mean (SD) | 7.6 (0.9)           | 8.4 (1.3)        | 8.1 (1.1)        |

Table 9 presents pairwise nABCD values with 95% bootstrap CIs for the three candidate effect modifiers.

**TABLE 9** Pairwise nABCD values with 95% bootstrap confidence intervals.

| Effect Modifier | Japan vs. US     | Japan vs. EU     | US vs. EU        |
|-----------------|------------------|------------------|------------------|
| Age             | 0.12 (0.07–0.18) | 0.08 (0.04–0.13) | 0.05 (0.02–0.09) |
| BMI             | 0.51 (0.44–0.58) | 0.38 (0.31–0.45) | 0.18 (0.12–0.24) |
| HbA1c           | 0.27 (0.20–0.34) | 0.19 (0.13–0.26) | 0.10 (0.05–0.16) |

#### 4.1.1 | Clinical Calibration

The nABCD values alone do not determine pooling decisions. We apply the clinical calibration procedure from Section 2.3 to translate distributional differences into potential treatment effect heterogeneity.

*Step 1: Estimate CATE sensitivity  $L$  for each EM.* From published subgroup analyses in diabetes trials, we adopt the following estimates for the Lipschitz constant of the CATE function:

- *Age:*  $L_{\text{age}} \approx 0.01\%$  HbA1c per year. Subgroup analyses typically show modest age–treatment interactions, with treatment effect varying by approximately 0.1% HbA1c per decade of age.
- *BMI:*  $L_{\text{BMI}} \approx 0.02\%$  HbA1c per  $\text{kg}/\text{m}^2$ . A regression analysis of DPP-4 inhibitor efficacy estimated a BMI coefficient of  $-0.02$  per  $\text{kg}/\text{m}^2$  ( $p = 0.024$ ) for the relationship between BMI and HbA1c reduction.<sup>15</sup>
- *HbA1c:*  $L_{\text{HbA1c}} \approx 0.3\%$  HbA1c per % HbA1c. Baseline HbA1c is the strongest effect modifier for most glucose-lowering therapies: meta-regression of 98 DPP-4 inhibitor trials estimated 0.4–0.5 percentage points greater HbA1c reduction per 1% higher baseline,<sup>16</sup> and a GLP-1 receptor agonist study estimated  $\beta = -0.31$  per % HbA1c after correcting for regression to the mean.<sup>17</sup>

*Step 2: Compute  $\Delta_{\text{max}}$  for the most discrepant region pair.* Table 10 presents the clinical calibration for the Japan–US comparison, which showed the largest nABCD values across all EMs.

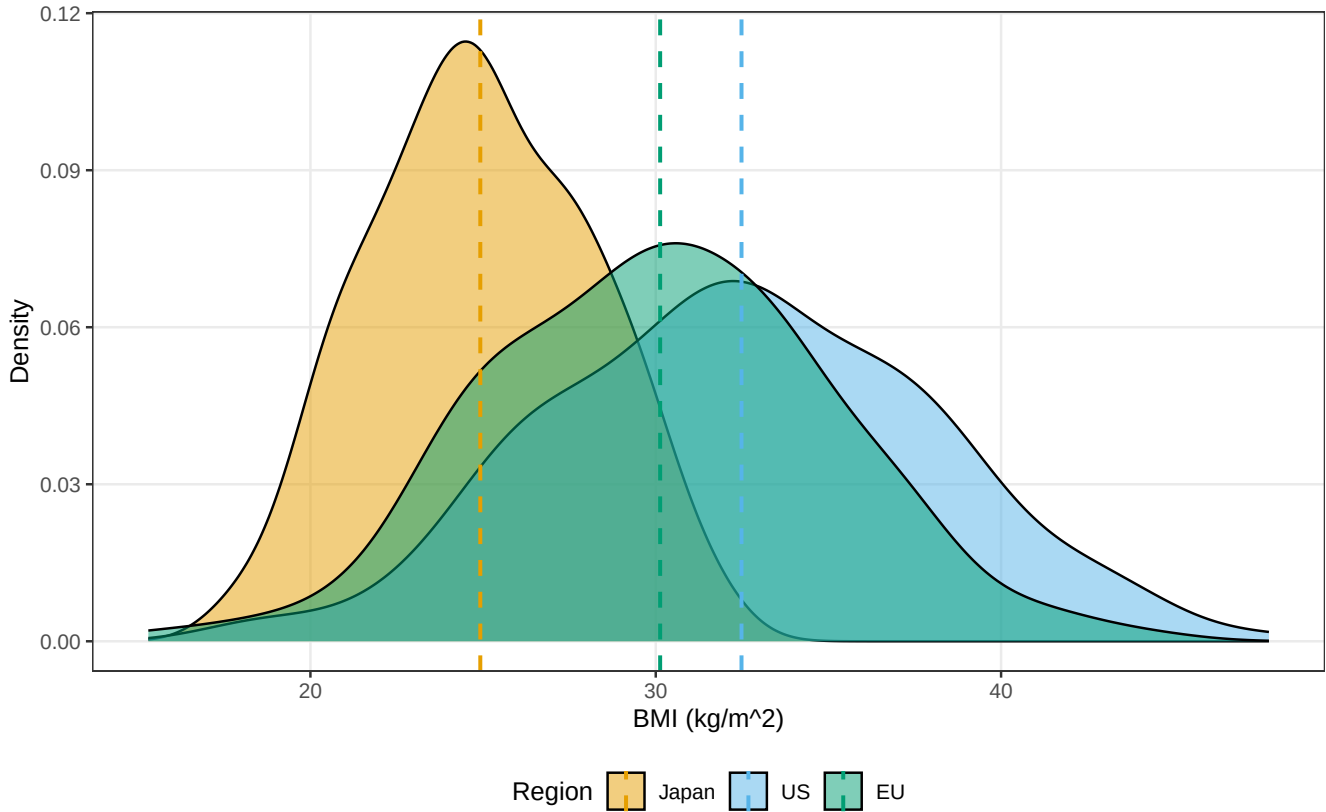
**TABLE 10** Clinical calibration of nABCD for the Japan–US comparison.

| EM    | nABCD | $L$  | $\text{IQR}_{\text{pooled}}$ | $\Delta_{\text{max}}$ | vs. $\Delta_{\text{clin}}$ |
|-------|-------|------|------------------------------|-----------------------|----------------------------|
| Age   | 0.12  | 0.01 | 14.2 yr                      | 0.03%                 | $\ll 0.4\%$                |
| BMI   | 0.51  | 0.02 | 7.8 $\text{kg}/\text{m}^2$   | 0.16%                 | $< 0.4\%$                  |
| HbA1c | 0.27  | 0.30 | 1.5%                         | 0.24%                 | $< 0.4\%$                  |

$\Delta_{\text{max}} = 2 \times L \times \text{IQR}_{\text{pooled}} \times \text{nABCD}$  (equation 7).  $\Delta_{\text{clin}} = 0.4\%$  HbA1c (non-inferiority margin).

*Step 3: Interpret  $\Delta_{\text{max}}$  in clinical context.*

The calibration reveals a nuanced picture that illustrates why clinical context, not nABCD alone, must drive interpretation:



**FIGURE 5** BMI distributions by region in the hypothetical diabetes MRCT. Japan shows substantially lower BMI compared to US and EU ( $nABCD = 0.51$  for Japan–US). However, the clinical impact depends on BMI’s role as an effect modifier, quantified through the heterogeneity bound  $\Delta_{\max}$ .

- *Age* ( $nABCD = 0.12$ ):  $\Delta_{\max} = 0.03\%$ , representing less than 4% of the overall treatment effect ( $-0.8\%$  HbA1c) and less than 8% of the non-inferiority margin. Age is a weak effect modifier for this drug class, and the distributional difference between regions translates to negligible potential heterogeneity.
- *BMI* ( $nABCD = 0.51$ ): Despite the large distributional difference,  $\Delta_{\max} = 0.16\%$ , or 20% of the overall treatment effect. BMI’s low CATE sensitivity ( $L = 0.02$ ) means that even substantial distributional differences have limited impact on treatment effect heterogeneity. Note that reference benchmarks alone would suggest “large difference”—an assessment that overlooks the weak relationship between BMI and treatment response for this drug class.
- *HbA1c* ( $nABCD = 0.27$ ):  $\Delta_{\max} = 0.24\%$ , reaching 30% of the overall treatment effect and 60% of the non-inferiority margin. Although the  $nABCD$  is smaller than for BMI, baseline HbA1c’s strong CATE sensitivity ( $L = 0.30$ ) means the moderate distributional difference carries greater clinical weight. This comparison warrants careful deliberation.

This example demonstrates that the clinical meaning of  $nABCD$  depends on the EM’s role as an effect modifier. A large  $nABCD$  for a weak EM (BMI,  $L = 0.02$ ) may be less consequential than a moderate  $nABCD$  for a strong EM (HbA1c,  $L = 0.30$ ). Because  $L$  is rarely known with precision, we recommend that practitioners report  $nABCD$  results across a range of plausible Lipschitz constants  $L$ , as the clinical interpretation depends critically on this parameter. Point estimates of  $\Delta_{\max}$  at a single  $L$  should be accompanied by sensitivity analysis showing how conclusions change across the plausible range. Table 11 illustrates this recommended practice for baseline HbA1c.

The sensitivity analysis provides trial designers and regulators with a transparent view: for this EM and region pair, at what level of CATE sensitivity does the distributional difference begin to matter clinically? This framing supports informed deliberation rather than a binary decision, consistent with the ICH E17 principle that “similar enough” requires contextual judgment.

The estimation precision from the simulation study (Section 3) propagates directly to the precision of  $\Delta_{\max}$ . For HbA1c ( $nABCD$  CI width  $\approx 0.14$  at  $n = 150$ – $200$ ), the CI width in  $\Delta_{\max}$  units is  $2 \times 0.30 \times 1.5 \times 0.14 = 0.13\%$  HbA1c—a level of

**TABLE 11** Sensitivity analysis:  $\Delta_{\max}$  as a function of assumed CATE sensitivity  $L$  for baseline HbA1c (Japan vs. US, nABCD = 0.27, IQR = 1.5%).

| $L$ (assumed) | $\Delta_{\max}$ | as % of treatment effect |
|---------------|-----------------|--------------------------|
| 0.10          | 0.08%           | 10%                      |
| 0.20          | 0.16%           | 20%                      |
| 0.30          | 0.24%           | 30%                      |
| 0.40          | 0.32%           | 40%                      |
| 0.50          | 0.41%           | 51%                      |

Overall treatment effect:  $-0.8\%$  HbA1c. The value  $L^* = \Delta_{\text{clin}}/(2 \cdot \text{IQR} \cdot \text{nABCD}) = 0.49$  represents the CATE sensitivity at which  $\Delta_{\max}$  equals the non-inferiority margin of  $0.4\%$ .

uncertainty small enough to support meaningful clinical calibration. For BMI, where  $L = 0.02$ , the same nABCD CI width translates to only  $2 \times 0.02 \times 7.8 \times 0.14 = 0.04\%$  HbA1c, demonstrating that estimation uncertainty in  $\Delta_{\max}$  scales with CATE sensitivity.

The analysis illustrates that nABCD serves as a measuring instrument for distributional distance, while the clinical judgment of whether that distance matters requires integration with domain knowledge about the EM's CATE sensitivity. The role of the statistician is to provide  $\Delta_{\max}$  and its uncertainty; the role of the clinical team is to evaluate that information against the therapeutic context.

## 5 | DISCUSSION

We developed and validated nABCD, a normalized metric for comparing effect modifier distributions in MRCTs. Our contributions include: (1) a principled metric combining Wasserstein-1 distance with IQR normalization; (2) a theoretical framework connecting nABCD to treatment effect heterogeneity through the heterogeneity bound; (3) a clinical calibration procedure that translates nABCD values into context-specific assessments using CATE sensitivity; and (4) rigorous validation through simulation.

The nABCD metric addresses limitations of current approaches. Compared to SMD, nABCD captures full distributional differences including variance and shape. Our simulation demonstrated that nABCD captured scale differences (S5), shape differences (S6), and skewness differences (S7) where SMD estimates remained near zero. Compared to the KS statistic, nABCD provides a direct connection to treatment effect heterogeneity through the heterogeneity bound (equation 7), enabling clinical calibration rather than purely statistical assessment. Compared to visual inspection, nABCD is objective and reproducible. Our approach is conceptually distinct from treatment effect consistency assessment methods<sup>2</sup> which evaluate whether observed treatment effects agree across regions; nABCD operates upstream, quantifying similarity in the effect modifier distributions that may drive such heterogeneity.

The choice of Wasserstein distance over information-theoretic divergences such as Kullback–Leibler (KL) divergence warrants discussion. While KL divergence is a natural measure of distributional difference in information theory, it has structural limitations for the present application. First,  $D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$ , meaning that comparing Region A to Region B yields a different value than comparing B to A—an asymmetry inconsistent with the symmetric nature of regional pooling decisions in MRCTs. Second, KL divergence can diverge to infinity when the support of  $Q$  does not cover the support of  $P$ , a common occurrence with empirical distributions that would require density estimation and bandwidth selection, introducing additional tuning parameters and complexity. Third, and most critically, KL divergence lacks the Lipschitz-bound property central to our framework: the Kantorovich–Rubinstein duality theorem ensures that Wasserstein distance bounds the difference in expectations of Lipschitz-continuous functions, enabling Proposition 2's heterogeneity bound. KL divergence has no analogous result linking it directly to treatment effect heterogeneity for Lipschitz-continuous CATE functions.<sup>18</sup> The Wasserstein distance is a true metric satisfying the triangle inequality and metrizing weak convergence,<sup>2</sup> making it the natural choice for our clinical calibration framework.

The clinical calibration procedure introduced in Section 2.3 represents a departure from fixed-threshold approaches. Cohen's  $d$  benchmarks (small = 0.2, medium = 0.5, large = 0.8) have been widely criticized for context-free application, and we deliberately avoid this pattern. Instead, the heterogeneity bound provides a principled mechanism to translate nABCD into clinically meaningful units. This approach respects the ICH E17 principle that “similar enough” is context-dependent: the same nABCD value may support pooling for one EM (where CATE sensitivity is low) but warrant separate analysis for another (where CATE sensitivity is high). Indeed, the application example revealed a *ranking reversal*: BMI had the largest

distributional difference ( $nABCD = 0.51$ ) yet the smallest clinical impact ( $\Delta_{\max} = 0.16\%$ ), while baseline HbA1c had a moderate distributional difference ( $nABCD = 0.27$ ) but the largest clinical impact ( $\Delta_{\max} = 0.24\%$ ) due to its 15-fold higher CATE sensitivity. This reversal underscores that distributional distance and clinical relevance are fundamentally different dimensions, and that  $nABCD$  magnitude labels (Table 2) describe distributional distance only, not clinical significance. This connection draws on transportability theory<sup>8,19</sup> to relate distributional distance to potential treatment effect heterogeneity.

Based on our findings, we offer the following recommendations for practitioners:

1. Compute  $nABCD$  with bootstrap CIs ( $n \geq 100$  per region) for each candidate EM across region pairs.
2. Translate  $nABCD$  into  $\Delta_{\max}$  and its CI on the clinical outcome scale using equation (7), incorporating prior knowledge or estimates of CATE sensitivity  $L$ .
3. Report  $\Delta_{\max}$  and its CI alongside the overall treatment effect, non-inferiority margin, or other clinically relevant benchmarks to support informed deliberation about pooling.
4. Conduct sensitivity analyses over plausible values of  $L$  when prior knowledge is uncertain (Table 11).
5. When  $L$  cannot be estimated, use the reference benchmarks in Table 2 as initial guidance, recognizing that clinical calibration provides a more rigorous basis for decision-making.

In practice, multiple candidate EMs may be evaluated simultaneously. The  $nABCD$  framework assesses each EM separately, and we recommend reporting  $\Delta_{\max}$  and its CI for every candidate EM across all region pairs. To arrive at an overall pooling recommendation, trial designers may adopt a conservative approach based on the maximum  $\Delta_{\max}$  across all EMs and region pairs, or they may take a totality-of-evidence approach in which the collection of  $\Delta_{\max}$  values, together with their uncertainties and the reliability of each  $L$  estimate, informs a holistic judgment. The choice between these strategies depends on the regulatory context: for applications where a formal pooling criterion is required, the maximum  $\Delta_{\max}$  provides a defensible worst-case assessment; for advisory settings where pooling decisions are informed by clinical deliberation, the full set of calibrated results offers a richer evidence base.

Several limitations should be acknowledged:

1. The current formulation applies to continuous EMs only; extensions to categorical or mixed-type EMs require further development.
2.  $nABCD$  evaluates each EM separately rather than jointly; a multivariate extension would address potential confounding among EMs.
3. Positive bias under the null at small sample sizes ( $n < 100$ ) can inflate  $nABCD$  estimates, and persistent negative bias at large true values (S4,  $nABCD = 0.372$ ) leads to progressive undercoverage (0.740 at  $n = 200$ ). These boundary effects are inherent to the non-negative, bounded nature of  $nABCD$  as a function of the Wasserstein distance. When the true value equals zero, the parameter lies on the boundary of the parameter space, where standard bootstrap consistency results may not apply;<sup>20</sup> we therefore do not report null-scenario coverage and recommend interpreting  $nABCD$  estimates near zero with caution. For non-null scenarios, the true value lies in the interior of the parameter space where standard bootstrap consistency holds, and our simulation confirms this with coverage of 0.87–0.98 at  $n \geq 100$  for most scenarios; notably, the new scenarios S7 (skew) and S8 (location + scale) achieved coverage between 0.93–0.95 across all sample sizes. Coverage can also exhibit non-monotonic behavior: for the shape scenario (S6), coverage was 0.576 at  $n = 50$ , 0.949 at  $n = 100$ , and 0.997 at  $n = 200$ , reflecting how the balance between bias magnitude and CI width shifts with sample size. We use the percentile bootstrap, which is first-order accurate; bias-corrected methods such as BCa showed inferior performance for this bounded statistic (Section 3.2), and the studentized bootstrap would require variance estimation for the Wasserstein distance ratio, which adds substantial complexity. We recommend  $n \geq 100$  per region for reliable point estimation and confidence intervals, and caution that coverage may degrade when the true  $nABCD$  exceeds approximately 0.3.
4. The clinical calibration requires estimation of CATE sensitivity  $L$ , which may not always be available from prior data. Reference benchmarks are provided as a fallback but should not replace context-specific calibration.
5. When multiple effect modifiers are relevant, the current framework evaluates each separately. Aggregation strategies for combining  $nABCD$  values across effect modifiers—such as weighted combinations reflecting clinical importance or multivariate extensions using higher-dimensional optimal transport—remain topics for future investigation.
6. As with any covariate-based approach,  $nABCD$  assesses similarity only with respect to measured effect modifiers. Unmeasured or unknown effect modifiers may contribute to treatment effect heterogeneity beyond what the metric captures.

Future work should address multivariate extensions, bias correction methods for small samples, and empirical calibration of CATE sensitivity parameters using real MRCT data. Methods for estimating  $L$  from historical trial databases would strengthen the clinical calibration framework. The nABCD framework could also be extended to longitudinal EM profiles, enabling dynamic assessment of distributional similarity over time.

An R package implementing the nABCD metric with bootstrap inference and the clinical calibration framework described herein is under development.

nABCD fills a methodological gap in ICH E17 implementation by providing a principled framework for assessing distributional similarity. Rather than reducing “similar enough” to a fixed threshold, the clinical calibration procedure translates distributional differences into context-specific assessments of potential treatment effect heterogeneity, enabling evidence-based and clinically grounded pooling decisions. Open-source R code is available to facilitate adoption.

## AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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## FINANCIAL DISCLOSURE

None reported.

## CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

## DATA AVAILABILITY STATEMENT

R code for computing nABCD and reproducing the simulation study is available at [repository URL].

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## SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website. Supporting materials include: complete R code for nABCD computation, simulation scripts, and additional figures for realistic clinical scenarios.

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## APPENDIX

### A PROOFS AND MATHEMATICAL DETAILS

#### A.1 Proof of Proposition 1

Non-negativity follows directly from the non-negativity of the Wasserstein-1 distance:  $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)| dx \geq 0$ , with equality if and only if  $F_1(x) = F_2(x)$  for all  $x$ . Since  $\text{IQR}_{\text{pooled}} > 0$  for non-degenerate distributions, the ratio  $\text{nABCD} = W_1/(2 \cdot \text{IQR}_{\text{pooled}}) \geq 0$ .

#### A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance  $W_1(\hat{F}_1, \hat{F}_2)$  is a consistent estimator of  $W_1(F_1, F_2)$ .<sup>11</sup> Combined with the consistency of the empirical IQR,  $\widehat{\text{nABCD}}$  is a consistent estimator of  $\text{nABCD}(F_1, F_2)$ .

In one dimension, the empirical  $W_1$  distance converges at rate  $O(n^{-1/2})$  without logarithmic correction.<sup>11</sup> For two-sample estimation with  $F_1 \neq F_2$  and finite second moments,  $\sqrt{n}(W_1(\hat{F}_{1,n_1}, \hat{F}_{2,n_2}) - W_1(F_1, F_2))$  converges in distribution to a mean-zero Gaussian, where  $n = n_1 + n_2$  with  $n_1/n \rightarrow \lambda \in (0, 1)$ . Since the empirical IQR also converges at rate  $O(n^{-1/2})$  under standard density conditions, the delta method applied to the ratio  $g(w, q) = w/(2q)$  yields asymptotic normality of  $\widehat{\text{nABCD}}$ :

$$\sqrt{n}(\widehat{\text{nABCD}} - \text{nABCD}(F_1, F_2)) \xrightarrow{d} N(0, \sigma_{\text{nABCD}}^2), \quad (\text{A1})$$

where  $\sigma_{\text{nABCD}}^2$  depends on the asymptotic variances and covariance of the  $W_1$  and IQR estimators. This result requires  $\text{nABCD} > 0$  (i.e.,  $F_1 \neq F_2$ ); at the boundary  $F_1 = F_2$ , the parameter lies on the edge of the parameter space and standard asymptotics do not apply (see Section 5).

The bootstrap provides valid inference for the Wasserstein distance under mild conditions.<sup>12</sup> The percentile bootstrap confidence interval achieves asymptotically correct coverage for non-degenerate distributions.

### B R CODE FOR NABCD COMPUTATION

**Listing 1** R function for computing nABCD with bootstrap confidence intervals.

```
compute_nABCD <- function(x1, x2) {
  pooled <- c(x1, x2)
  iqr_pooled <- IQR(pooled)
  if (iqr_pooled == 0) return(NA)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
      F2(all_vals[-length(all_vals)])))
  w1 / (2 * iqr_pooled)
}
```



```
nABCD_bootstrap <- function(x1, x2,
  B = 2000, conf = 0.95) {
  obs <- compute_nABCD(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_nABCD(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
    ci_lower = ci[1], ci_upper = ci[2])
}
```